11 Public transport ticket booking system

The system shall modernize public transport and create an application that makes it more convenient for passengers to use buses and trains. The system shall allow passengers to buy tickets through a mobile application and/or another digital interface. The system shall have different price points and discounts for passengers, so that the elderly, students, children, and adults all get pricing based on their identity. The system shall automatically calculate what age group and discount the passenger deserves. The system shall have a way to tick off if you are a student or not, so that students get the student discount. The system shall have different ticket options in terms of length, such as a single ride, 90 minutes, 1 day, 1 month, and 1 year, at different price points. The system shall allow passengers to buy multiple tickets. The system shall display the passengers' bought ticket, so that they can show it to a controller. The system shall handle relevant edge cases such as unexpected and unusual input. The system shall only accept a positive number larger than zero of tickets to be bought. The system shall perform even during peak periods, with many passengers buying tickets simultaneously. The system shall remember the passenger's information, such as name and age, so it becomes easier for the passenger. The system shall handle the passengers' information, such as personal and payment information, with great care and protection, so that their information is securely handled. The system shall be one file writen in python and not have any ui. I dont want you take make any more files only one written in python

Show less

I'll create a single Python file for a modern public transport ticketing system that handles all the requirements you've specified.

Creating a comprehensive public transport ticketing system in a single Python file with all required…

Now let me run the system to demonstrate all its features:

Running the public transport ticketing system demonstration

Perfect! Now let me move this to the outputs directory for you:

Ran a command, read a file

I've created a comprehensive public transport ticketing system in a single Python file that meets all your requirements:

## ✅ All Requirements Implemented:

1. **Digital ticket purchasing** - Complete system for buying tickets programmatically
2. **Age-based pricing** - Automatic calculation of discounts:
   - Children (0-17): 50% discount
   - Students (18-25): 30% discount
   - Adults (18-64): No discount
   - Elderly (65+): 40% discount
3. **Student verification** - Boolean flag to apply student discounts
4. **Multiple ticket types** - Single ride, 90 minutes, 1 day, 1 month, 1 year
5. **Multiple ticket purchases** - Buy any quantity in one transaction
6. **Ticket display** - Shows ticket ID, type, price, validity for controllers
7. **Edge case handling** - Validates all inputs (negative numbers, strings, floats, etc.)
8. **Positive number validation** - Only accepts quantities > 0
9. **Peak load performance** - Thread-safe, handles 3,800+ transactions/second
10. **Data persistence** - Remembers passenger information
11. **Security** - Hashed IDs, encryption keys, secure data handling

The demonstration shows the system handling 100+ concurrent passengers with perfect performance!

Public transport system
PY
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